The Domino Half Adder: what the simulation showed

East bay 2: the Domino Half Adder · 20 October 2026

Every junction was tested on its own before the machine used it, with `tests/junctions.lua` (run from `tests/`; its header lists the settings). Dominoes are 48 x 24 x 7.5 mm, 8 g, friction 0.4, no bounce, at 2400 steps a second, unless a line says otherwise.

Lines

- **Spacing.** A straight line of 30 fell completely at every spacing from 1.6 to 4.6 cm. The wave runs at 113 cm/s at 1.6 cm, 99 at 2.8 and 63 at 4.6. The machine uses 2.6 cm.
- **Turns** of radius 5, 8 and 12 cm through 90 degrees: all fell.
- **Pushing the first domino:** setting a spin about its centre does nothing; most of it is lost against the table at once. It has to be turned about its front foot (spin plus the matching forward speed), as a finger pushes it.

Junctions

- **Fork:** the trunk's last domino strikes the two branches' first dominoes, 1.0 or 1.4 cm either side of its line, at 20 or 30 degrees. All four settings worked.
- **Merge:** two feeders at 15 to 35 degrees, ending 0.3 to 0.7 cm either side of the outgoing line, each set off it alone. All worked; 0.5 cm and 30 degrees are used.
- At 1.4 cm a single feeder only twisted the outgoing domino, so the merge acted as an AND.
- A feeder's last domino must stand exactly where it should. Spacing a feeder to fit its length, instead of stopping it up to a spacing short, fixed a wave that died at the first domino it struck.
- **Tap** (a branch leaving a line that carries on, 1.4 cm aside, at 25 degrees):
- **with the wave:** taken.
- **against the wave:** not taken. Only the branch's first domino falls (20 and 30 degrees both).

This is what makes the XOR. - **Head-on:** two waves on one line, from both ends: they meet and stop. - **Back-running:** a wave reaching the far end of the collision line also runs back up the other input's feeder. It is harmless: it passes the carry's tap the wrong way (not taken) and stops at that input's switch gap.

The cut (inhibitor)

A line arriving square to another, its last domino 3.6 cm from that line's centre, knocks dominoes out of it sideways. The gap stops the other line's wave if the cut comes first.

- **Distance:** at 3.2, 3.6 and 4.0 cm the cut worked; at 2.4 cm the main line ran on.
- **Head start:** the inhibitor needs about 4 frames (70 ms). 1 frame was not enough; 4 to 12 always worked at 3.6 and 4.0 cm.

- **Aim:** aimed between two dominoes, it can slip through the gap between them without touching either; there is only 0.27 cm to spare. So the carry line is spaced wide where it is cut: three dominoes 4.25 cm apart, the inhibitor aimed at the middle one. One domino out leaves 8.5 cm, which no wave jumps.
- At 3.8 cm the wave still got across at friction 0.3: a domino slid over the fallen one.
- At 4.5 cm the wave didn't cross even uncut at friction 0.3.
- 4.1 to 4.4 cm worked at both 0.3 and 0.5.

Switches

A single missing domino is not a gap: the next one is 4.45 cm away, within reach of a 4.8 cm domino. Two missing leave 7.8 cm, which stops a wave. The two dominoes taken out must be laid well aside, across the line. Laid 7 cm along it, they bridged the gap.

The template

The template's pocket walls (1.6 mm high, 1.2 mm wide, 0.4 mm clearance) are in the simulation as fixed boxes.

At first a wall of one pocket stood inside a neighbour's pocket at the crowded junctions. That domino was pushed over at the start: SUM fell with both inputs 0. The walls are now cut where they would stand in another pocket, as on the print (frames minus pockets). That leaves 1076 wall boxes round 268 dominoes.

The whole machine

Before the tell-tales were added (the half adder alone, and the full machine started from GO through its switches): - **Half adder:** all three non-zero sums right at friction 0.3, 0.35, 0.4, 0.45 and 0.5. At each friction, four placings with every domino misplaced at random (1 mm, 2 degrees): 20 of 20 right. - **Full machine, from GO through the switches, with the template:** all four sums right at friction 0.3, 0.4 and 0.5. At each friction, three placings with every domino misplaced within its pocket (0.3 mm, 1.5 degrees): 9 of 9 right.

The finished exhibit (`domino-adder.lua`'s own tour of all four sums, with the tell-tales and the switches laid aside), at friction 0.3, 0.4 and 0.5, three placings each with every domino misplaced within its pocket (`DA_JIT=0.03`): 9 tours of 9, 36 sums of 36, right.

Two faults were found on the way: - **Switches:** the removed dominoes were first laid 7 to 10 cm along the line instead of across it, and bridged the gap; the waves ran on with both inputs 0. - **Paths:** the parts file was looked for by `$PWD`, which is wrong when bpp has moved into the script's folder; the real working directory is used now.

Resting

Once the hand has nothing to do and nothing has moved for half a second, the Domino Half Adder stops stepping until there is something to do again. Stepped at 2400 steps a second, a machine at rest costs as much as one running, and its parts never quite stop (their sleeping is turned off, as it must be while they work): what was left once it had settled was up to about 0.15 cm/s and 0.07 rad/s (fallen dominoes rocking against each other). "Not moving" is set just above that: 0.3 cm/s and 0.1 rad/s. Resting can't change a result: woken, it carries on from exactly the state it stopped in, speeds and all. It rests only between runs, so the runs themselves are as before. Over a 50 s run, 165 s of computing became 88 s.